

When Numerical Structure Reads the Recording Pipeline: A Boundary for Factorization-Based Data Analysis

Abstract

Digit-based forensic tests are increasingly run against published official statistics, where an extreme test statistic may reflect how the values were recorded rather than how they were generated. In a real case, a standard election-forensics last-digit test fires catastrophically ($\chi^2_9 = 12,684$) on the Census Bureau’s citizen-voting-age-population tabulation—a modeled series published on a five-unit rounding grid—while genuine enumerated returns pass. The test cannot distinguish fabrication from a documented publication rule. We present a small set of integer-arithmetic *recording-grid coordinates* that identify such alarms as grid-compatible—and, where the recording rule is documented, attribute them to it—demonstrated here and in a second case localizing a survey publication grid to its margin-of-error column. The design follows from the paper’s central result: within the recorded-data classes tested, and under stated recording assumptions, factorization *shape*—what a value’s prime factorization looks like once prime names are discarded—retains only the value’s magnitude and its most recent recording or quantizing operation, not its deeper generating history. Across roughly 19,000 integer channels, all apparent shape structure reduces to four drivers (magnitude, decimal rounding, small-prime composition, residue quantization); grouped held-out analysis shows the apparent shape geography carries essentially no process information beyond recording variables. The boundary is mechanistic—one added unit destroys the multiplicative-construction signal—and powered: a host-matched quotient-level alternative is recovered at a 5% mixture in six of six channels. Exact, unrounded multiplicative integers remain a positive regime. The practical warning for official-statistics users, forensic analysts, and auditors: a detector firing on a recorded number does not establish that it read the underlying activity; it may have read the last recording operation. All confirmatory claims were prospectively specified in internally frozen analysis plans; three missed their frozen rules and are reported.

Keywords: data quality; data provenance; digit analysis; rounding; heaping; quantization; prime factorization; election forensics; pre-registration

Statement of Significance. Analysts screen published statistics—population tabulations, employment series, survey margins of error—with digit-based tests for signs of fabrication or error. This paper documents, on real federal statistical products, how badly such a screen can mislead: a standard forensic test declares an extreme anomaly on an ordinary Census

tabulation whose only peculiarity is that the agency rounds published values to the nearest five. We provide a simple set of arithmetic measurements, computed from the published integers alone, that identify when an alarm is explained by such a recording rule, and we establish—with prospectively specified analyses across roughly 19,000 data columns—why this works: in the data classes tested, after ordinary recording (summing, counting, rounding), the arithmetic fine structure of a number retains the recording rule and essentially nothing else about how the quantity arose. The result protects data users from a documented class of false accusations, and warns that statistical extremity alone does not license conclusions about data integrity.

1 A false alarm on a published statistic

An analyst screening turnout-related integer series for irregularities will sooner or later apply a last-digit uniformity test (Beber and Scacco 2012) to the U.S. Census Bureau’s citizen-voting-age-population (CVAP) tabulation—a natural denominator for turnout (U.S. Census Bureau 2023c). On county-level values of at least 1000 (the test’s standard magnitude screen), the test statistic is $\chi^2_9 = 12,684$ (p below reporting precision): among the most extreme digit anomalies one can observe on a published series. Run beside it, genuine certified election returns pass comfortably—North Carolina precinct returns and nationwide county returns for two presidential elections give $p \geq 0.16$ (North Carolina State Board of Elections 2024; McGovern 2024). Figure 1 shows what the test saw: nearly all CVAP values end in 0 or 5.

The alarm is real; the natural interpretation is wrong. CVAP is not an enumerated count. It is an American Community Survey special tabulation—a *modeled estimate*—and the Bureau publishes it rounded to the nearest five (U.S. Census Bureau 2023c,d). A five-grid forces terminal digits into $\{0, 5\}$; the test fires on the publication rule. The test statistic cannot know that: last-digit uniformity is a valid fabrication screen only under assumptions about how the values were generated *and recorded*, and a publication grid violates them innocently. The coordinates in Figure 1(b), computed from the published integers alone, supply the missing step: the residue distribution of the CVAP values modulo 25 exceeds a magnitude-matched uniform reference by +0.77 in total variation—the arithmetic fingerprint of a five-grid—while every enumerated series carries none; the Bureau’s documented rounding rule then completes the attribution. The correct disposition is “recording artifact,” not “irregularity.”

Three scope statements, made once. CVAP is a modeled published series, not a vote count. The analysis throughout is methodological—about recording rules and the reading of digit-forensic alarms—and makes no claim about the legitimacy of any election. And the case does not show that forensic alarms are generally innocent; it shows that an alarm cannot be interpreted until the recording process has been considered.

2 Research question and contributions

The CVAP case poses the question this paper answers: *when a statistical detector fires on a recorded number, what did it actually read—the generating process, or the way the values*

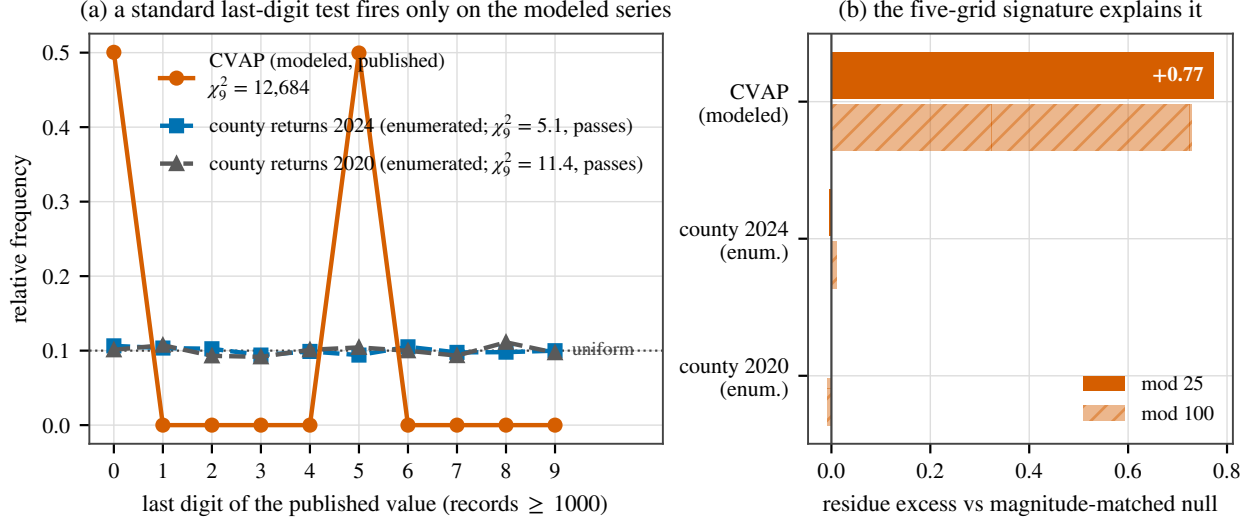


Figure 1: The CVAP false alarm and its attribution. (a) Last-digit distributions on records ≥ 1000 : the modeled CVAP series concentrates on terminal 0 and 5 and fires the Beber–Scacco test at $\chi^2_9 = 12,684$; enumerated county returns are uniform and pass. (b) The recording-grid coordinates identify the alarm as grid-compatible: CVAP’s residue distribution modulo 25 (and 100) departs from the magnitude-matched uniform reference by +0.77 (+0.73), the direct signature of rounding to the nearest five; every enumerated series is at zero. The Bureau’s documented nearest-five rule completes the attribution.

were recorded?

We answer it for one family of representations: integer factorization. Every integer’s prime factorization is a complete invariant, and it is natural to hope that a fine arithmetic lens might read provenance—whether a value was built by multiplication, counting, or estimation—directly off stored data. Data-quality practice already reads coarser projections of the same digits: leading-digit conformance tests (Benford 1938; Hill 1995; Nigrini 2012), last-digit and duplication screens (Beber and Scacco 2012), heaping indices (Myers 1940; Whipple 1919; A’Hearn et al. 2009), and price-clustering diagnostics (Harris 1991). The paper makes three contributions, and no more:

1. **A conditional boundary** on what factorization shape retains in recorded administrative data: within the recorded-data classes tested, and under the recording assumptions stated in Section 3, factorization shape reflects magnitude and the most recent recording or quantizing operation—not deeper generating history (Sections 4–5).
2. **A powered, mechanistically supported explanation** for the boundary: one additive unit destroys the multiplicative-construction signal, and the pipeline demonstrably recovers a matched surviving alternative at a 5% mixture, so the negative is not underpowered (Sections 6–7).
3. **A narrow recording-grid attribution instrument**, predicted by the boundary (if the grid is what survives, measure the grid), demonstrated in two real cases—the CVAP

false alarm above and an ACS margin-of-error localization—that prevents a demonstrated class of grid-induced forensic false interpretations (the gate’s own output is “grid-compatible”; attribution is claimed only where the recording rule is documented) (Section 8).

The instrument’s demonstrated real-data role is attribution in two cases, not general fabrication detection; on real returns it does not outperform established digit tests as a detector, and we say so where the comparison is made. Digit-based tests are treated throughout as complementary projections with their own assumptions, not as inferior methods.

3 A minimal model of representation and recording

Four objects carry the whole argument; everything else (algorithms, null construction, finite-size detail) is in the supplement.

Factorization shape: the L-profile. For an integer $n = \prod_{i=1}^r p_i^{a_i}$, each prime-power part contributes $a_i \ln p_i$ of the value’s log-magnitude. Normalizing and sorting gives the *L-profile*

$$\ell_j = \frac{a_{(j)} \ln p_{(j)}}{\ln n}, \quad \ell_1 \geq \ell_2 \geq \cdots \geq \ell_r, \quad \sum_j \ell_j = 1, \quad (1)$$

from which we read $L_1 = \ell_1$, $L_3 = \ell_3$, $\text{Tail} = 1 - \ell_1 - \ell_2$, and the concentration $H_2 = \sum_j \ell_j^2$. The profile is a *shape* object: it records how log-magnitude is spread across multiplicative parts. Every claim in this paper about “shape” concerns functionals of the log-share multiset $\{a_i \ln p_i\}$ —blind to prime *names* given their sizes, though not to the sizes themselves (small primes carry small log-shares, which is what makes driver 3 of Section 5 visible). Shape is deliberately distinguished from *named-prime* instruments: a readout keyed to particular primes (say, the factor 5 a rounding rule injects) is a different projection with a different boundary, and the attribution instrument below is prime-aware by design.

The magnitude-conditioned baseline. The reference for shape is the behavior of uniform random integers. Their normalized factor sizes converge to the Poisson–Dirichlet law (Billingsley 1972; Kingman 1975; Arratia et al. 2000), with $\mathbb{E}[L_1] \rightarrow 0.6243$, but convergence is far too slow for real data: $\mathbb{E}[L_1 \mid d]$ falls from 0.952 for 1-digit integers to 0.633 at 18 digits (a Dickman-type computation (Dickman 1930)). Magnitude alone therefore moves the profile more than any effect studied below, and all structural claims are residuals against this *magnitude-conditioned baseline* $\mathbb{E}[\cdot \mid d]$, with d the digit length. The channel-level summary is a ratio of averages,

$$k_{\text{eff}} = \frac{\sum_d w_d \mathbb{E}[H_2 \mid d]}{\overline{H_2}}, \quad (2)$$

with w_d the channel’s digit histogram and $\overline{H_2}$ its mean concentration over records (records with fewer than three prime-power parts contribute their exact H_2 ; the structural-zero treatment of L_3 is in Section 4). It equals 1 for generic integers, exceeds 1 when multiplicative energy spreads across more parts than chance, and falls below 1 for concentration.

The recording-grid coordinates. Where the L-profile is blind to prime names, the attribution instrument reads exactly the primes and residues a recording rule injects: the *grid mass* ρ (the log-share of a value carried by the primes 2 and 5, split into its 2- and 5-components), the *grid depth* (the power of ten dividing the value), *grid conformance* (the fraction of records on the channel’s modal grid), and *residue quantization* Q_m (the total-variation distance of values modulo m from uniform, scored against a magnitude-matched reference, at moduli such as 25, 49, 47, 43). All are computed from stored integers alone; a units change (dollars to cents) shifts ρ predictably and leaves the residual coordinates invariant. The supplement gives the full definitions and the channel-level algorithm.

The recording model, and the shape window. A recorded integer is modeled as

$$n = g \cdot q, \quad g = 2^s 5^t m_0, \quad q = \text{round}(x/g), \quad (3)$$

where x is the latent quantity, g is the recording grid (a decimal grid, optionally times a non-decimal tick m_0), and q is the stored quotient. A publication grid is one kind of recording rule; the model covers any rule whose last operation multiplies a generic quotient by a fixed factor. Under one modeling assumption—*local flatness*, that the pre-rounding density of x/g is roughly constant across one grid cell—the quotient q is, to first order, a generic integer: its factorization carries no imprint of the process that produced x . Equation (3) then fixes the division of labor. The shape of n is the shape of $g \cdot q$; since q is generic, all non-magnitude shape structure comes from g . The grid mass and depth read the decimal part of g ; the residue coordinates read a non-decimal tick m_0 that leaves no trace in terminal digits. We call the multiplicative structure a value retains its *shape window*: **the accumulated multiplicative history since the last additive or quantizing operation applied to the value**. For an exact product never subsequently summed or rounded, the window contains the whole construction. For an ordinary recorded value, the model predicts the window contains the grid and nothing else.

Scope, stated once. Two qualifications govern everything that follows. First, the genericity of q rests on local flatness and on the near-independence of the factorizations of consecutive integers; the latter is a heuristic—the exact statement is a Chowla-type correlation problem, open in general (Tao and Teräväinen 2019)—so the boundary is a *conditional empirical claim*, not a theorem. Second, the completeness of the four-driver reduction below is an empirical finding in the tested corpus, consistent with the model’s prediction, not a proof that no fifth driver exists. The claims are scoped to shape functionals of the log-share multiset $\{a_i \ln p_i\}$ (blind to prime names given their sizes), on the recorded-data classes tested, under model (3).

4 The apparent geography

Suppose an analyst suspects that integer fields of different provenance carry different multiplicative character, and maps a large corpus of fields by their factorization shape. We ran exactly this. The corpus is 18,951 integer

Table 1: Corpus composition (frozen post-quarantine set: 18,951 channels, 367 dataset-families, 18 domain labels).

domain	channels	families	domain	channels	families
network_telemetry	14,929	6	seismology	84	34
procurement	1,812	8	payments_fraud	77	11
census	1,105	26	sports	77	13
unresolved	159	79	real_estate	55	12
equity_markets	146	103	food_nutrition	27	10
campaign_finance	137	3	genomics	13	11
health	115	8	crime	11	3
admin_codes	102	3	accounting	10	4
corporate_financials	90	32	taxi_trips	2	1

channels (one integer-valued column of one dataset, deduplicated by content fingerprint) in 367 dataset-families across 18 domains—equity markets, census, network telemetry, procurement, seismology, real estate, health, and others—with $\sim 9.6 \times 10^8$ exact factorizations (Table 1; construction details in the supplement; the “roughly 19,000” used elsewhere rounds this frozen count, and the supplement’s §S4 footnote reconciles the two related profile-set counts). The **unresolved** row collects channels whose domain the frozen labeling rules could not assign; they are excluded from every domain-labeled analysis. Channel counts are dominated by a few high-volume administrative sources, a property the reader needs when judging the scope of every claim below.

The map looks like a discovery. In the raw (L_1, Tail) plane the 367 families form three clean, reproducible clusters (silhouette 0.719); in the raw (L_1, L_3) plane of Figure 2(a) the channels trace a structured band with a prominent arm. A shape-based provenance taxonomy—“these fields are multiplicative, those are counts, those are prices”—appears within reach.

Figure 2(a) already contains the refutation, because every comparator on the map is matched. The uniform-integer baseline traces exactly the band’s spine: position along the band is magnitude. The arm is decimal rounding: exact products relocated onto the arm by a single grid rounding sit among the recorded channels. Additive controls are indistinguishable from the baseline. The only points far from the band are exact multiplicative constructions that no recording operation has touched. (One representational caveat is marked explicitly: 37.5% of sampled records have fewer than three prime-power parts, so per-record L_3 is a structural zero, not a small continuous value; channels dominated by such records are faceted separately, and the supplement gives the per-record analysis.)

Two quantitative results close the case against the taxonomy reading. First, the frozen corpus analyses showed the three clusters are ordered segments of a single variable—the grid mass ρ —and dissolve when the decimal primes are divided out (cluster survival Jaccard 0.39–0.43); a supervised “domain signal” on monetary channels reverses sign under a units change (cents versus dollars), exposing it as a scale artifact; and the cleaned space is a one-dimensional magnitude continuum (intrinsic dimension 1.16 by the TwoNN estimator (Facco et al. 2017)) with no domain islands. Second—new in this version, and exploratory—a

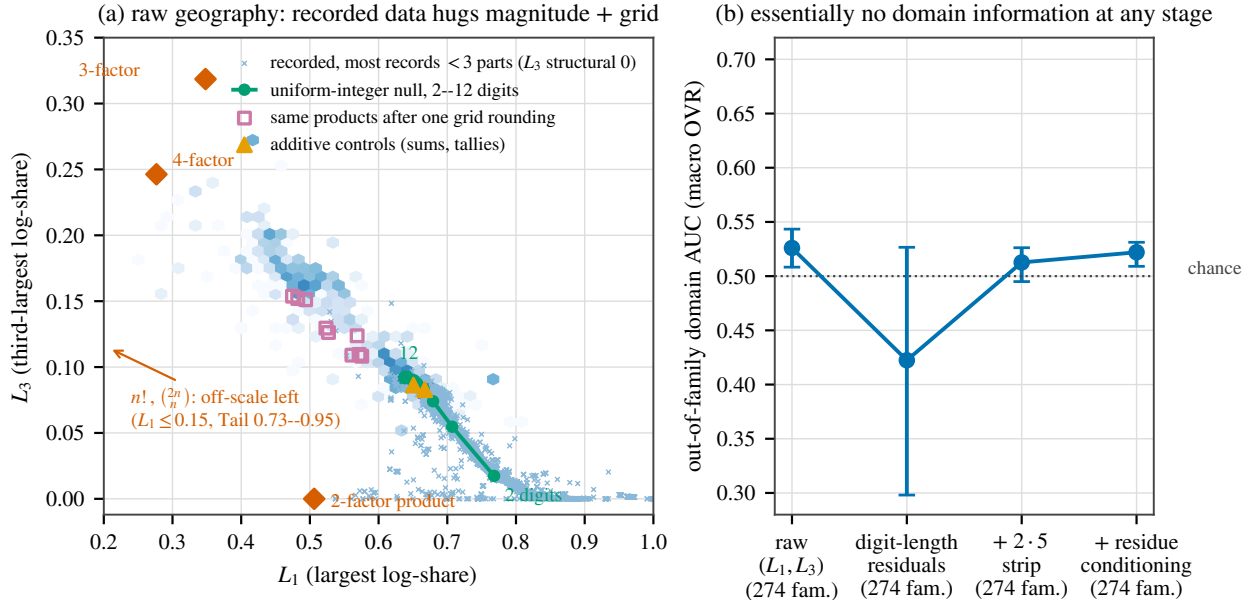


Figure 2: The apparent geography and what it maps. (a) Channel-level (L_1, L_3) plane: recorded channels (blue density; crosses mark channels whose records mostly have fewer than three prime-power parts, for which L_3 is structurally zero) hug the magnitude-conditioned baseline curve (green: uniform-integer pseudo-channels, 2–12 digits) extended by a decimal-rounding arm. Exact multiplicative constructions (red diamonds) sit far off the band—and land back on it after a single decimal-grid rounding (open squares). Additive controls (sums, tallies; orange) sit exactly on the baseline. (b) Out-of-family domain information in the coordinates at each control stage (grouped 5-fold by dataset-family, macro one-vs-rest AUC, family-bootstrap 90% intervals): the map’s visible structure carries essentially no cross-source process information even before controls. Exploratory analysis (Build geo15).

grouped transfer test asks whether the coordinates carry *any* domain information across independent dataset-families: predicting a channel’s domain from its raw (L_1, L_3) position, with entire families held out, achieves macro AUC 0.526 (90% CI [0.508, 0.543]: above 0.5, but weak and practically negligible), and no stage restores it (Figure 2(b)). A positive control calibrates the design’s sensitivity: the identical family-grouped pipeline reads synthetic recording differences at AUC 0.81 for a grid-mass shift of +0.020 and 0.99 at +0.066, with a family-permutation null centered at 0.485 (supplement §S4). The transfer information the real coordinates carry is therefore an order of magnitude below what a modest recording difference produces. The apparent geography was never a usable process taxonomy, even before controls: it is a map of magnitude and recording, mistaken for one because magnitude and recording conventions correlate with domains within any fixed corpus.

5 The four-driver reduction

The claim that the map is “magnitude and recording” is made precise by an explicit reduction. The full distributional deviation of each channel’s profile from the magnitude-

Table 2: The four-driver reduction: each driver of raw shape structure, its effect, and the operator that removes it. After all four, no shape structure remains (the surviving candidate collapses under its own control, Figure 3).

driver	effect	removal operator
1. magnitude (digit length)	dominant; raw map tracks the baseline curve	magnitude-conditioned baseline $\mathbb{E}[\cdot \mid d]$
2. decimal rounding (2·5)	the entire raw arm; Tail excess up to +0.11	divide out $2^{v_2}5^{v_5}$, re-baseline the core
3. small-prime composition (3, 7; 11, 13 via the ≤ 13 strip; 2, 5 belong to driver 2)	25–50% of the de-rounded overshoot	strip primes $\leq 7, \leq 13$
4. residue quantization (ticks, grids)	incremental $\Delta R^2 = +0.043$; partial correlation $r = 0.488$	residue scores Q_m , negative-controlled
(<i>surviving candidate</i>)	concentration excess -0.0184	core-magnitude re-indexing $\rightarrow -0.0001$

conditioned baseline—measured by Wasserstein and energy distances, so the whole shape is compared, not two summary moments—is attributed hierarchically to four drivers, each with a removal operator (Table 2).

Three of the four removals are arithmetic and uncontroversial; the residue driver is validated by planted controls at $\sim 9\text{--}30\times$ separation (Figure 3(b))—this driver is the digit-preference (heaping) phenomenon of demography (Myers 1940; A’Hearn et al. 2009) and the price-clustering phenomenon of market microstructure (Harris 1991), measured in factorization coordinates rather than digit space, and it sits in the coarse-data tradition of Heitjan and Rubin (1991), Manski and Molinari (2010), and, for heaped survey income specifically, Drechsler and Kiesl (2016). (Driver 3 is probed at strip cuts ≤ 7 and ≤ 13 ; the core-magnitude re-indexing below standardizes on the $c = 7$ cut, and the ≤ 13 variant agrees—supplement §S2.) The reduction is also source-robust: removing network telemetry, procurement, or census from the corpus wholesale, or weighting families equally, leaves the driver sequence and the collapse below unchanged (telemetry-excluded core residual -0.0001 , 90% family-bootstrap CI $[-0.0004, +0.0004]$; per-domain family counts and all variants in supplement §S3).

One signal initially survived every strip: the deep cores of rounded domains looked *more* concentrated than chance (excess -0.0184). This is the linchpin the boundary turns on, and it dissolves under its own control. The subtlety is the reference: rounded values carry more small-prime mass, lose more magnitude when stripped, and so have smaller cores—and smaller integers are intrinsically more concentrated. Re-indexing the null by the *core’s* magnitude rather than the original value’s (the *core-magnitude baseline*; algorithm in the supplement) collapses the excess to -0.0001 (90% CI $[-0.0004, +0.0004]$): 99.5% of the “surviving signal” was reference misspecification, a finding we present as the correction it is. A mechanism check agrees: genuine concentration requires exponent stacking, which observational cores essentially never show (1.9% of records, versus 100% for genuine smooth

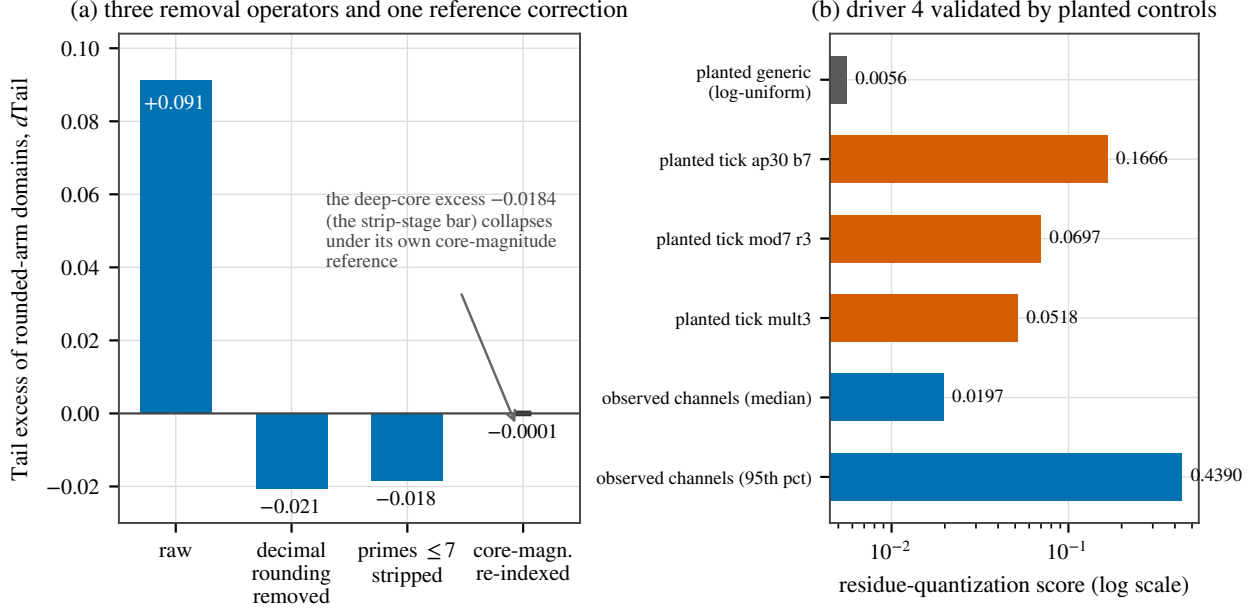


Figure 3: Driver removal. (a) Tail excess of the frozen arm domains at family-equal weighting (the same estimand as the leave-one-source-out analysis, supplement §S3): raw (driver 1 removed by the baseline), after dividing out decimal rounding (driver 2), after stripping primes ≤ 7 (driver 3)—this strip-stage value *is* the deep-core concentration excess -0.0184 —and after core-magnitude re-indexing, the reference correction under which it collapses to -0.0001 (90% family-bootstrap CI $[-0.0004, +0.0004]$). (b) Driver 4 validated by planted negative controls: channels quantized onto ticks score $10\text{--}30\times$ above planted generic (log-uniform) channels on the residue score; observed channels span the same range.

products).

The reduction has an out-of-sample check on grid specificity: binary-aligned storage allocations show a factor-2 excess ($+0.72$) with factor-5 mass at baseline (-0.007)—a dissociation no decimal convention produces—while unaligned file sizes are null. The shape residue reads whichever grid the recording system actually used.

6 Mechanism: why recording empties the shape window

The reduction is not a corpus accident; it is what the recording model predicts, and its sharpest consequence is verified directly.

Factorization is discontinuous under addition: n and $n + 1$ are coprime, and the joint multiplicative structure of consecutive integers is the territory of Chowla-type correlation problems, where only partial results exist (Tao and Teräväinen 2019). Recording, by its nature, performs additive operations—summing, counting, rounding (projection onto a grid, which replaces a value’s factorization with the grid’s times a fresh quotient). Under local flatness, the quotient is generic, and the recorded value’s shape window is empty except for

the grid.

Figure 4(a) shows the verification. On balanced composite products with matched magnitudes and moments, shape-based detection reads AUC 0.86 with no grid; a decimal grid of width 2 degrades it to 0.75; width 25–100 reaches chance. The minimal additive perturbation $n \mapsto n+1$ —one unit, a change of relative size 10^{-12} for these values—collapses detection to 0.44, below chance (a parity echo). **One additive unit destroys the multiplicative-construction signal.** The same operation relocates exact products onto the baseline in the geography of Figure 2(a): in the exploratory stress analysis, a three-factor product at $(L_1, L_3) = (0.35, 0.32)$ moves to $(0.62, 0.10)$ under $n+1$ —the twelve-digit baseline point is $(0.64, 0.09)$.

The local-flatness assumption itself has named failure regimes, mapped by a stress simulation (supplement §S2). When the latent density’s curvature is comparable to the grid width (curvature-to-grid ratio $r \lesssim 2$), or for sparse zero-inflated counts on coarse grids, the recorded quotient is *not* generic: it inherits narrow support and biases the shape residual toward spurious structure in either k_{eff} direction, never toward masking a real signal (k_{eff} of the quotient 1.34 at $r = 0.5$ in the exponential case and 0.94 in the zero-inflated count case; within ± 0.02 of 1 by $r \geq 5$). Benchmarked estimates are the third named regime (the retired whisper of the supplement’s misfire ledger). Practically: the boundary and the instrument are calibrated for series whose values spread over several grid cells; on narrow-support or mostly-zero series the magnitude-matched reference itself is the first thing to check.

We are careful about epistemic register here, once: the coprimality of n and $n+1$ is exact algebra; the degradation curves are empirical findings; the genericity of quotients is a heuristic consistent with them; and the independence of consecutive factorizations in full generality is an open mathematical question. The boundary claim inherits the weakest of these—it is a conditional empirical boundary, not a theorem.

7 The null is powered: a quotient-level spike-in

A negative result is only as strong as its demonstrated sensitivity, and the demonstration must not be circular. An earlier design planted exact composite products into rounded channels; those injections are off-grid values in on-grid hosts, so their recovery could reflect grid drift rather than multiplicative detection (they shift the mixture’s grid coordinates at $|z|$ up to 10^7). That design is demoted to illustration.

The de-circularized design plants signal exactly where the model says it could survive: in the quotient. Each injected record keeps its host record’s grid part and magnitude and replaces only the quotient with a composite product; the most adversarial tier also matches the host’s joint small-prime composition, leaving composite structure in the deep core as the *only* difference from a generic quotient. Detection uses the stripped-core effective factor count $k_{\text{eff}}^{\text{core}}$ (Eq. (2) computed on the deep core against the core-magnitude baseline), with a mandatory confound check that the mixture’s eleven grid coordinates stay within bootstrap noise of the host ($\max |z| < 3$). Full tiers, thresholds, and per-channel results are in the supplement.

The result (Figure 4(b)): the host-matched tier is recovered in six of six channels at a 5% mixture (confound-clean in five; the one flag is *channel resonance*—residue-class con-

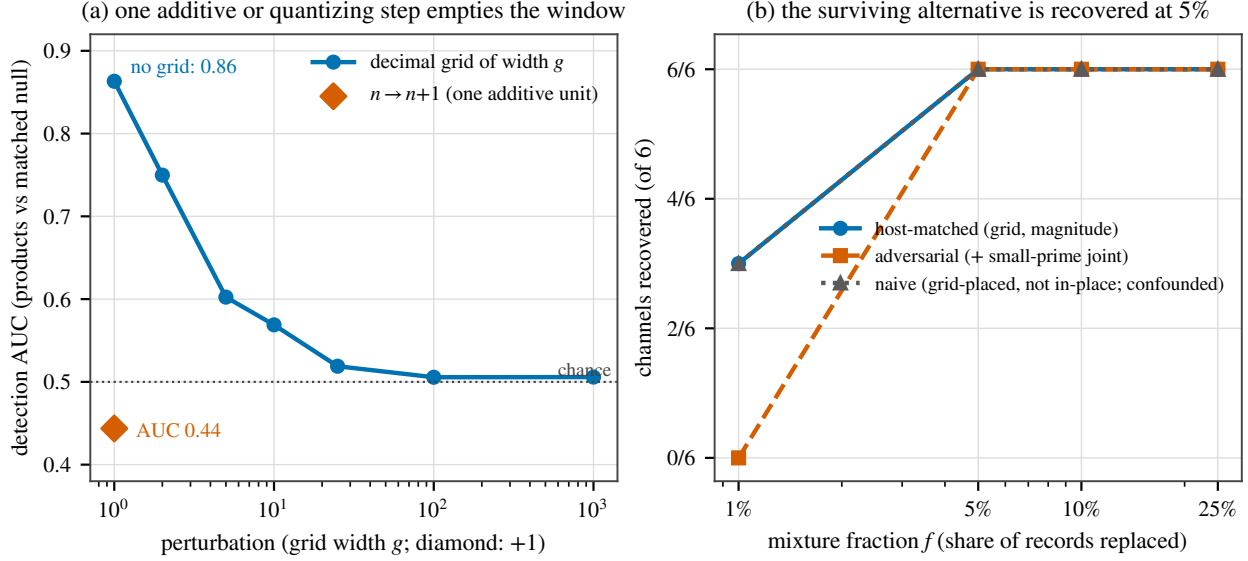


Figure 4: Destruction and power. (a) Detection AUC for exact composite products under recording operations: grid coarsening degrades detection to chance by width ~ 25 – 100 ; one additive unit ($n \rightarrow n+1$, diamond) collapses it below chance. (b) Quotient-level spike-in: fraction of six real host channels recovered versus mixture fraction, by injection tier. Both matched tiers are recovered 6/6 at $f = 0.05$; the confound check ($\max |z| < 3$) is clean in 6/6 channels for the adversarial tier and 5/6 for the host-matched tier (Section 7).

centration intrinsic to the host’s own values—which the injections dilute, not a grid leak), and the adversarial tier is recovered six of six at 5%, confound-clean in all six. The unit of detection deserves emphasis: “recovered at 5%” means the *channel-level* core statistic moves past the channel’s own 95th-percentile noise band when one record in twenty is replaced—population-level change detection, not identification of individual records. The negative is therefore not underpowered *against the model-implied quotient-level multiplicative alternative in the tested host channels*: six amount-kind channels here, and—in an exploratory v10.3 extension with its plan frozen in advance—three count-kind hosts (telemetry byte counts, the only count channels meeting the frozen magnitude screen), where the same in-place injection is again recovered 3/3 at a 5% mixture. The extension’s frozen rule reads, verbatim: “count-host recovery HOLDS iff $\geq$ half of hosts recovered at $f \leq 0.10$ with clean confound” (clean = $\max |z| < 3$); recovery is met, the confound condition is not, so the extension does not qualify as clean recovery. In the count hosts the mixture-level *residue* coordinates drift below the host ($|z|$ 3.3–5.2): the injections dilute the hosts’ strong intrinsic resonance, the same mechanism as the one flagged amount host, while the decimal-grid coordinates stay within noise—so the flags mark dilution of host structure, not grid leakage (supplement §S5). When this pipeline finds no residual in real channels, the evidence supports absence of quotient-level multiplicative-history signal in the tested classes, not absence of sensitivity.

Pre-registration covered this and every other confirmatory analysis: twenty decision rules were frozen in configuration files before computation; seventeen met their bars and three misfired. None of the misfires touches a load-bearing conclusion—one retired a candidate

whisper signal exactly as its replicate-or-retire rule required, and two exposed miscalibrated secondary statistics whose mechanical causes are instructive—and all three are reported in full in the supplement rather than repaired.

8 The instrument: measure the grid, not the shape

The boundary dictates the instrument. If the only multiplicative history a recorded value retains is its grid, then a usable provenance readout should measure the grid directly—and no per-record shape functional can do better, since every one of them inherits the brittleness of Section 6. The recording-grid coordinates of Section 3 are that readout. Their demonstrated contribution is precisely bounded: *two real-data attribution instances, one constructed detection case, and calibration contrasts*—not a general fabrication detector.

Attribution instance 1: the CVAP false alarm. Section 1 is the flagship. The coordinates attribute the $\chi^2 = 12,684$ alarm to a five-grid: grid mass 0.275 against 0.12 for enumerated votes, grid depth 0.55, and residue excess +0.77 modulo 25 (+0.73 modulo 100) against ≈ 0 for every enumerated series (Figure 1(b)). The gate’s own output is “grid-compatible”; it becomes an *attribution* here because the Bureau’s documentation states the nearest-five rounding rule (U.S. Census Bureau 2023d), closing the loop from signature to mechanism. We claim an existence proof of the attribution capability, not a generally validated procedure.

Attribution instance 2: the ACS margin-of-error localization. The American Community Survey provides the case where the instrument found structure a digit analyst would not have tested (U.S. Census Bureau 2023b). The pre-registered expectation was that ACS county *point estimates* would be grid-marked relative to the decennial census. They are not (grid mass 0.107 versus 0.112; the frozen detection bar was missed and is reported as missed). The publication grid lives instead in the *margin-of-error* column: grid mass 0.183 (90% CI [0.151, 0.216]) against 0.112 (Figure 5). We have not located Census documentation stating a margin-of-error rounding rule explicitly, so this instance is “consistent with a publication grid”—a localization, not a documentation-closed attribution; CVAP remains the documented instance (the ACS technical documentation U.S. Census Bureau 2023a describes the products generally). The grid mark is subtle—partial rounding, visible to the CI-based grid-mass comparison but below the coarse residue threshold of the validation library in supplement §S10. The two real instances differ usefully in kind: CVAP is a false-alarm filter on the column an analyst does test; ACS is a localizer pointing to a column nobody tests.

Bounded detection, and honest benchmarking. When the recording mechanism is a *non-decimal* grid, the residue coordinates can detect what terminal-digit tests cannot: on a constructed contrast—real values re-quantized onto a 47-tick versus a 43-tick, both invisible modulo 100—the digit battery sits at or below AUC 0.53 while the residue coordinates read the tick at 1.00. That is a scope demonstration on constructed data, not real-world validation; no natural non-decimal channel appeared in our corpus. On real returns the instrument adds no detection value: read two-sided, a last-two-digit drift test reaches AUC

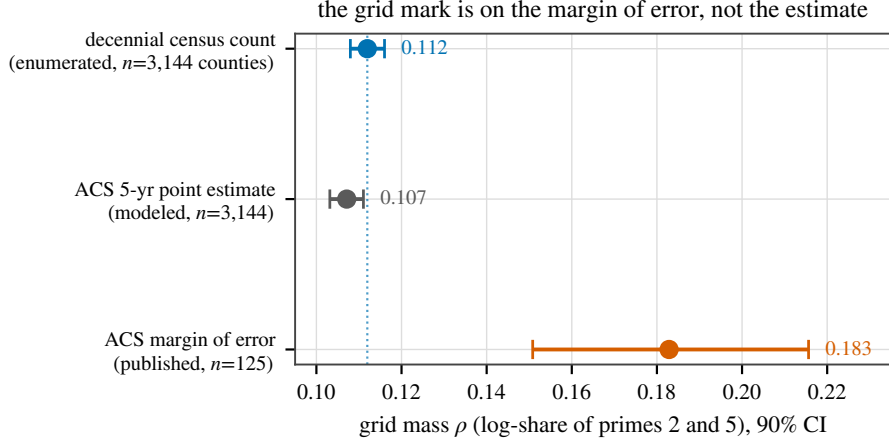


Figure 5: ACS margin-of-error localization: grid mass with 90% bootstrap intervals for county-level cross-sections. The modeled point estimate is not grid-marked relative to the enumerated count (the pre-registered expectation that it would be was wrong, and is reported as a miss); the published margin of error is. The margin-of-error cross-section yields 125 usable county values under the frozen extraction screens; the estimate columns yield 3,144.

1.00 on the fabrication contrast and dominates the grid coordinates (0.68–0.83). We report the adverse comparison because omitting it would be spin; the instrument’s real-data role is attribution.

Calibration on two further quantities. Employment and votes calibrate the two axes on series of known provenance. Survey employment estimates against the administrative near-census of the same quantity: multiplicative axis null at every vintage (k_{eff} 0.98–1.00), survey hundreds-grid cleanly separated on the grid axis (grid mass 0.474 versus 0.102) (U.S. Bureau of Labor Statistics 2024a,b). Enumerated vote counts: k_{eff} 0.98–1.03 at precinct, county, and state aggregation—additive tallies, as expected—with residuals at trace scale (+0.004 to +0.016) appearing exactly where the prospectively specified small-count caveat applies and absent elsewhere. With population (ACS/decennial), the boundary is confirmed on three field quantities (employment, population, votes); the fourth, byte counts, is consistent—null at k_{eff} 0.996 across 6,399 telemetry count channels, with the count-host power extension exploratory and its confound bar not met (Section 7).

How a practitioner uses it. The demonstrated pattern is a two-step gate on a flagged field. First, run whatever digit test raised the concern. If it fires, compute the grid coordinates on the flagged partition against a clean reference. Elevated grid mass, non-trivial grid depth, or residue excess at a modulus matching a plausible tick attribute the alarm to a recording rule; their absence means the anomaly is *not* attributable to a grid this instrument can see, and substantive investigation should proceed. The gate’s output is always “grid-compatible” or “not attributable to a visible grid”—never a verdict on the data. A worked adversarial example makes this operational (supplement §S10): values *fabricated* from a smooth generator and then published on a five-grid fire the last-digit test ($\chi^2_9 \approx 11,800$) and

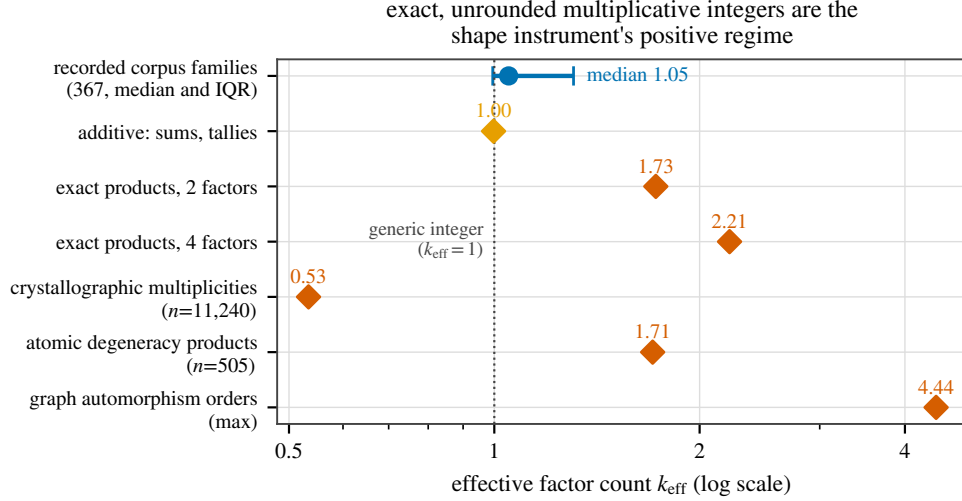


Figure 6: The positive regime: effective factor count by class. Recorded corpus families and additive constructions sit near the generic value $k_{\text{eff}} = 1$; exact multiplicative constructions and measured multiplicative fields depart in both directions.

the gate correctly reports grid-compatible—because the alarm *is* caused by the grid. The gate attributes the alarm, not the data: it filters false positives; it never certifies innocence.

9 The complementary regime: exact integers

The boundary is a statement about recorded data, not about the instrument’s intrinsic power, and the contrast deserves one compact section. Where integers are exact, unrounded, produced by multiplication, and never subsequently severed by addition or quantization, the shape window contains the whole construction, and the profile is a working two-sided detector (Figure 6). On synthetic ground truth matched in magnitude and moments: sums and Poisson tallies read $k_{\text{eff}} = 1.00$ to two decimals (0.997–0.998), while products of k factors separate at AUC 0.86–0.96. On genuine structured products with closed-form factorizations (factorials, binomials, finite-group orders; magnitudes to 10^{7263}), all classes clear the pre-registered AUC bar of 0.85. On measured fields the detector reads both directions: crystallographic multiplicity products at $k_{\text{eff}} = 0.534 \pm 0.091$ ($n = 11,240$), atomic degeneracy products at 1.707 ± 0.557 ($n = 505$), graph automorphism orders (exact, to $10^{17,265}$) at k_{eff} up to 4.44—while real additive fields sit at 0.996. This regime occurs by construction in mathematics and the exact sciences and almost nowhere in the administrative data classes tested here; the full catalogue, the balanced-pair edge case, and the exact-arithmetic computation notes are in the supplement.

Table 3: The central warning, in usable form.

“Significant departure” does not by itself mean:	
fabrication or misconduct	the rule that recorded the data can produce the same signature
risk or process change	the recording rule may have changed, not the process
data-quality failure	a documented publication grid is working as designed
For a grid-attributed alarm, a valid conclusion is:	
“The observed anomaly is consistent with, and attributable to, a documented recording grid.”	
It is not:	
“The underlying records were fabricated,” or—equally—“the underlying records are genuine.”	
Attribution of the alarm is not certification of the data.	

10 Implications for data quality, forensic analysis, and audit

The general lesson generalizes beyond factorization: a diagnostic reads only what its projection encodes, and for recorded administrative integers—in the classes tested here—the projection’s window typically ends at the last recording operation. We translate this into reporting practice.

A reported numerical anomaly on recorded data should state: the *unit of analysis* (record, channel, series); the *scored population* (which records survived screens); the *representation* the detector reads (digits, residues, factorization shape); the *baseline* against which “anomalous” is defined (uniformity, magnitude-conditioned null, historical control); any *known recording or publication rule* for the series; whether the result is *detector-only* (a statistic fired), *statistically calibrated* (fired against a validated null), or *construction-attributed* (the responsible mechanism was identified); the *compatible alternative mechanisms*; and, explicitly, *what conclusion is licensed and what is not*.

Table 3 is the compact form, and supplement §S10 turns it into an operational protocol: the default modulus set and modal-grid recipe, *z*-score and magnitude-matched-reference conventions, a multiplicity note, stop/go criteria, the worked CVAP report with every field filled, the fabricated-and-gridded adversarial example, and a ground-truth validation library of seven grid-truth series (three documented real, four constructed) and six non-grid series (sensitivity 6/7, specificity 5/6, both misses mechanistically explained: subtle partial rounding sits below the coarse threshold, and small-magnitude series can false-flag—the same failure regime named in Section 6). In the CVAP case the licensed conclusion is exactly the first: the alarm is attributable to the documented five-grid. Nothing about the underlying tabulation’s accuracy follows, in either direction. The same discipline applies to the affirmative use of any factorization-shape score on recorded data: a channel that looks “anomalously composite” or “anomalously concentrated” against a naive reference is, in the classes tested

here, exhibiting its magnitude and its recording rule—as the deep-core episode of Section 5 shows, even a careful analysis can manufacture a survivor by mis-indexing the reference.

For provenance practice, explicit lineage metadata—the W3C PROV model (Moreau and Missier 2013) and the provenance literature more broadly (Buneman et al. 2001; Simmhan et al. 2005; Cheney et al. 2009)—are strictly better than inference from values where they exist. The grid coordinates are a single-column profiling primitive in the tradition of automated data profiling (Abedjan et al. 2015; Huhtala et al. 1999; Papenbrock et al. 2015; Hellerstein 2008), for the common case where lineage is absent and the only witness is the stored integers. The digit-forensics literature has long known that its tests can mislead when recording assumptions fail (Deckert et al. 2011; Mebane 2011; Klimek et al. 2012; Lacasa and Fernández-Gracia 2019), and integer-percentage fingerprints of falsification have their own electoral literature (Kobak et al. 2016); Beber and Scacco (2012) note that rounding must be excluded for the last-digit test to be informative, but give no method. The attribution gate is such a method for one class of mechanism.

11 Limitations and conclusion

Limitations, stated precisely and once. The boundary is a conditional empirical claim under model (3), not a theorem; its completeness is an empirical finding in one large but partly internal corpus dominated by a few administrative sources. It concerns shape functionals of the log-share multiset (blind to prime names given their sizes); named-prime instruments are a different projection with a different boundary, and nothing here says prime-specific arithmetic rules cannot survive recording. The powered null covers quotient-level composite structure and balanced-pair mimics in six amount-kind host channels. The attribution capability is demonstrated in two real instances, not validated as a general method; the pre-specified gate is validated only on a small ground-truth library (seven grid-truth series—three documented real, four constructed—and six non-grid series; one miss each way, both explained); the non-decimal detection case is constructed, and locating a natural coprime-tick channel is future work. The exact-integer results are positive contrasts at demonstration scale. Field results cover three statistical systems in one country. The new L1/L3 geography analyses are exploratory (grouped-holdout design, family-level blocking, thresholds declared in advance and labeled as project rules), and three pre-registered predictions failed their frozen rules and are reported in the supplement. The electoral material remains strictly methodological, as stated in Section 1.

Conclusion. Factorization shape reads the accumulated multiplicative history since the last additive or quantizing operation. For exact multiplicative integers, that window can contain the whole construction; for ordinary recorded administrative values—in the classes tested, under the stated recording model—it is generally reduced to magnitude and the recording grid. A factorization-shape geography of recorded integers is therefore usually a map of the recording pipeline, and the practical instrument the boundary licenses reads the grid directly—which is exactly what turns a catastrophic-looking digit alarm on a modeled population series into what it was all along: a documented rounding rule, working as designed. The profile reads the multiplicative history available after recording; for ordinary

administrative records, that history is often only the grid.

Data availability

All analyzed series are public federal and state statistical products: BLS Current Employment Statistics (archival real-time vintages) (U.S. Bureau of Labor Statistics 2024a) and Quarterly Census of Employment and Wages (U.S. Bureau of Labor Statistics 2024b); Census Bureau ACS 5-year estimates (U.S. Census Bureau 2023b), decennial and Population Estimates Program products, and the CVAP special tabulation (U.S. Census Bureau 2023c); North Carolina State Board of Elections precinct returns (North Carolina State Board of Elections 2024) (the certified-canvass primary source) and a community compilation of county-level results assembled from state canvass pages and media tabulations (McGovern 2024), used as a calibration comparator; and, for the exact-integer regime, the Crystallography Open Database (Gražulis et al. 2024), the NIST Atomic Spectra Database (NIST 2024), and the SNAP network collection (Leskovec and Krevl 2014). The frozen configuration files, decision-rule manifest, per-claim manifest (`claim_manifest.csv`, mapping every numeric claim to script, output, and SHA-256 hash), and hash-verified outputs are archived as a Zenodo deposit (DOI reserved; an anonymized reviewer link is provided at submission), to be made public on publication. The “public products” sentence above covers the field-calibration series; the corpus’s family-level access status (which source files are public, which are hash-referenced local snapshots) is inventoried in the deposit. The analysis uses only published aggregates, so the PRICSSA checklist for complex-sample microdata analyses (Seidenberg et al. 2023) does not apply.

Code availability

The analysis and figure code—extraction, exact factorization, grid coordinates, spike-in, the exploratory geography build (`stage22_geo15.py`, with unit tests), and all figure scripts—is version-controlled in a git repository [repository link at publication]. The exact frozen state used for this manuscript (code, configuration files, the decision-rule manifest, the claim manifest, and hash-verified outputs) is archived as a Zenodo deposit (DOI reserved; anonymized reviewer link provided at submission), with a worked end-to-end verification of the CVAP result in the archive README. Repository and deposit will be made public on publication.

Prospective specification

Decision rules for all confirmatory analyses were prospectively specified in internally frozen configuration files before computation (20 rules); this is internal freezing with hash integrity, not external registry timestamping (supplement §S6). Three registered predictions failed their frozen rules; they are reported, with mechanical causes, in the supplement (§S6). The L1/L3 geography analyses of Sections 4 and 6 labeled “exploratory” were not covered by a frozen rule and are reported under the analysis plan frozen in `run_config_geo15.json` before results were computed.

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